

Lethal autoimmune myocarditis in interferon-gamma receptor-deficient mice: enhanced disease severity by impaired inducible nitric oxide synthase induction.

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Evidence abstract

BACKGROUND Interferon-gamma (IFN-gamma) is an essential cytokine in the regulation of inflammatory responses in autoimmune diseases.

Little is known about its role in inflammatory heart disease.

METHODS AND RESULTS We showed that IFN-gamma receptor-deficient mice (IFN-gammaR^{-/-}) on a BALB/c background immunized with a peptide derived from cardiac alpha-myosin heavy chain develop severe myocarditis with high mortality.

Although myocarditis subsided in wild-type mice after 3 weeks, IFN-gammaR^{-/-} mice showed persistent disease.

The persistent inflammation was accompanied by vigorous in vitro CD4 T-cell responses and impaired inducible nitric oxide synthase expression, together with evidence of impaired nitric oxide production in IFN-gammaR^{-/-} hearts.

Treatment of wild-type mice with the nitric oxide synthetase inhibitor

N:-nitro-l-arginine-methyl-ester enhanced in vitro CD4 T-cell proliferation and prevented healing of myocarditis.

CONCLUSIONS Our data provide evidence that IFN-gamma protects mice from lethal autoimmune myocarditis by inducing the expression of inducible nitric oxide synthase followed by the downregulation of T-cell responses.

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